**AAE 5200 Experimental Aerodynamics**
**Class + Lab -- 3 Credits**

## 1 Course Information/Description

In this course students learn about the theory and application of experimental techniques and instrumentation for aerodynamics and fluid mechanics. Other topics covered include computer data acquisition, wind tunnels, force balances, flow visualization, pressure probes, hot-wire anemometry, laser Doppler velocimetry, and turbulence measurements.

### Learning Resources and Reference Text

There is no textbook. However, students will be expected to study reference materials in order to understand how to use and operate the experimental apparatus. The following are listed as recommended references for the course, and should thus be available in the bookstores. If you plan to keep working in the area it would be worth purchasing them, as opposed to borrowing them. The first text is also on reserve in the Engineering Library.

1. Low-Speed Wind Tunnel Testing, J.B. Barlow, W.H. Rae, Jr., and Alan Pope, Third Edition, Wiley, New York, 1999. Contains a lot of standard information on low-speed wind tunnels and also some information about instrumentation.

2. Fluid Mechanics Measurements, edited by Richard Goldstein, Jr. Various editions are available. Latest is published by Taylor and Francis, 1996. Used copies may also be available, my own copy is published by Hemisphere in 1983. Best known single reference for common instrumentation.

3. Writing Styles and Standards in Undergraduate Reports, 2nd edition, by Sheldon Jeter and Jeffrey Donnell, College Publishing, 2011. One of many books on writing reports. You might prefer A Scientific Approach to Writing for Engineers and Scientists, by Robert Berger, Wiley and IEEE Press, 2014.

## 2 Assignments / Topics Covered

• Lab 1 – Turbulence and Wakes: Calibrate hot wire in wake tunnel. Place wire in cylinder wake, study signal acquisition effects such as aliasing. Sample data, compute power spectra of turbulence under different conditions. Measure profiles of mean velocity and fluctuations at different downstream locations, for a cylinder and an airfoil. Airfoil may be placed at small angle of attack. Integrate wake profiles to get drag, look at effect of downstream distance on the result. Compare results to results from the literature. Two 4-hour sessions in successive weeks.

• Lab 2 – Forebodies at High Angle of Attack: Place cone in water tunnel at high angle of attack. Visualize vortex shedding with dye. Vary flow speed and angle of attack. Operate Laser Doppler Velocimeter. Measure profiles of velocity across vortex, mean and rms. Compare results to results from literature. Two 4-hour sessions in successive weeks.

• Lab 3 – Shock/Boundary-Layer Interaction at Mach 2.0: Run empty tunnel and get pressure distribution. Put wedge in blank in supersonic tunnel, using Mach-2.0 nozzle. Repeat, check pressure distribution. Connect Kulite pressure sensors to pressure taps on centerline near compression corner

(disconnect from tunnel and plug). Measure mean and fluctuations in pressure distribution at different tunnel pressures. Use oil flow on floor of tunnel, photograph to look for separation. Obtain Schlieren images. Compare. Two 4-hour sessions in successive weeks.

• Project: Instructor will suggest some possible topics, with the assistance of other interested faculty. Student-initiated topics are welcomed. Students form small groups, propose topics for approval by instructor, research background information, design experiments, develop apparatus and perform measurements, analyze the data, iterate the process, and write up the results in a final report.

### 2.1 3 Pre-Planned Lab Reports

A lab report documenting the methods and results for the first three pre-planned labs is required. The length of the report shall be no longer than 25 pages.

### 2.2 Final Project

The final project involves a white paper, literature review, proposal, and final report documenting the procedures, methods, and results.

## 3 Course Learning Objectives

Through taking this course, the students should be able to:

1. Improve their ability to plan, execute, analyze and report experimental measurements in aerodynamics.
2. Improve their understanding of common instrumentation in aerodynamics, and common types of experimental facilities.

## 4 Evaluation

Three preplanned labs, 45% total. Project, 50%. Recitations, 5%. A grading rubric for the first three lab reports can be found on the website for the course.

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| 3 Pre-planned Lab Reports | 45% |
| Final Project | 50% |
| Recitation Participation | 5% |